Course Code: CSE2202

Course Title: Algorithms Lab

Section: 4

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Insertion Sort

```c
#include<stdio.h>
void  Insertion_sort (int arr[] , int n)
{
    int step = 0 , index;
    for (int i=0 ; i<n ; i++)
    {
        index = arr[i] ;
        int j = i - 1 ;
        step++ ;
        while (j >= 0 && arr[j] < index)
        {
            step++ ;
            arr[j+1] = arr[j] ;
            step++ ;
            j-- ;
        }
        arr[j + 1] = index ;
        step++ ;
    }
    for (int i=0 ; i<n ; i++)
    {
        printf("%d " , arr[i] ) ;
    }
```

```
    printf("\nThe step is : %d\n" , step) ;
}
int main()
{
    int n = 9 , array1[9] , array2[9] , array3[9] ;
    printf("Enter the 9 digit of my id number : ") ;
    for (int i=0 ; i<n ; i++)
    {
        scanf("%d" , &array1[i] ) ;
    }
    for (int i=0 ; i<n ; i++)
    {
        array2[i] = array1[i] ;
        array3[i] = array1[i] ;
    }
    for (int i=0 ; i < n-1 ; i++)
    {
        for (int j=i+1 ; j<n ; j++)
        {
            if (array2[i] > array2[j] )
            {
                int temp = array2[i] ;
                array2[i] = array2[j] ;
                array2[j] = temp ;
```

```
            array2[j] = temp ;
        }
    }
}
for (int i=0 ; i<n-1 ; i++)
{
    for (int j = i+1 ; j<n ; j++)
    {
        if ( array3[i] < array3[j] )
        {
            int temp = array3[i] ;
            array3[i] = array3[j] ;
            array3[j] = temp ;
        }
    }
}
printf ("My id is : ") ;
Insertion_sort (array1 , n) ;
printf ("Ascending order : \n") ;
Insertion_sort (array2 , n) ;
printf ("Descending order : \n") ;
Insertion_sort (array3 , n) ;
return 0 ;
}
```

OUTPUT

```
Enter the 9 digit of my id number : 2 4 2 0 1 4 1 6 5
My id is : 6 5 4 4 2 2 1 1 0
The step is : 60
Ascending order :
6 5 4 4 2 2 1 1 0
The step is : 84
Descending order :
6 5 4 4 2 2 1 1 0
The step is : 18

Process returned 0 (0x0)   execution time : 7.625 s
Press any key to continue.
```

Selection Sort

insertion_sort.c *Selection_Sort.c

```c
#include<stdio.h>
void Selection_sort (int arr[] , int n , int decending)
{
    int max_index , step = 0 ;
    for (int i = 0 ; i < n-1 ; i++)
    {
        max_index = i ;
        for (int j = i+1 ; j<n ; j++)
        {
            step++ ;
            if (decending)
            {
                if (arr[j] > arr[max_index] )
                {
                    max_index = j ;
                }
            }
            else
            {
                if (arr[j] < arr[max_index] )
                {
                    max_index = j ;
                }
            }
```

```
            }
        }
        if (max_index != i)
        {
            int temp = arr[i];
            arr[i] = arr[max_index];
            arr[max_index] = temp;
            step++;
        }
    }
    for (int i=0 ; i<n ; i++)
    {
        printf ("%d " , arr[i] ) ;
    }
    printf ("\nThe step is : %d\n" , step ) ;
}
int main()
{
    int n = 9 , array1[9] , array2[9] , array3[9] ;
    printf("Enter the 9 digit of my id number : ") ;
    for (int i=0 ; i<n ; i++)
    {
        scanf("%d" , &array1[i] ) ;
    }
```

ere Insertion_sort.c *Selection_Sort.c

```
47      }
48      for (int i=0 ; i<n ; i++)
49      {
50          array2[i] = array1[i] ;
51          array3[i] = array1[i] ;
52      }
53      for (int i=0 ; i < n-1 ; i++)
54      {
55          for (int j=i+1 ; j<n ; j++)
56          {
57              if (array2[i] > array2[j] )
58              {
59                  int temp = array2[i] ;
60                  array2[i] = array2[j] ;
61                  array2[j] = temp ;
62              }
63          }
64      }
65      for (int i=0 ; i<n-1 ; i++)
66      {
67          for (int j = i+1 ; j<n ; j++)
68          {
69              if ( array3[i] < array3[j] )
70              {
```

```
            if ( array3[i] < array3[j] )
            {
                int temp = array3[i] ;
                array3[i] = array3[j] ;
                array3[j] = temp ;
            }
        }
    }
    printf ("My id is : ") ;
    Selection_sort(array1 , n , 1) ;
    printf ("Ascending order : \n") ;
    Selection_sort(array2 , n , 0) ;
    printf ("Descending order : \n") ;
    Selection_sort(array3 , n , 1) ;
    return 0 ;
}
```

OUTPUT

```
E:\iTubeGo\Code\Selection_S

Enter the 9 digit of my id number : 2 4 2 0 1 4 1 6 5
My id is : 6 5 4 4 2 2 1 1 0
The step is : 42
Ascending order :
0 1 1 2 2 4 4 5 6
The step is : 36
Descending order :
6 5 4 4 2 2 1 1 0
The step is : 36

Process returned 0 (0x0)   execution time : 6.786 s
Press any key to continue.
```

Quick Sort

```c
#include<stdio.h>
int step = 0 ;
void swap (int *a , int *b)
{
    int temp = *a ;
    *a = *b ;
    *b = temp ;
    step++ ;
}
int partition (int arr[] , int low , int high)
{
    int pivot = arr[high] ;
    step++ ;
    int i = low - 1 ;
    for(int j = low ; j < high ; j++)
    {
        step++ ;
        if (arr[j] > pivot)
        {
            i++ ;
            swap( &arr[i] , &arr[j] ) ;
        }
    }
    swap ( &arr[i + 1] , &arr[high] ) ;
```

```c
    swap ( &arr[i + 1] , &arr[high] ) ;
    return i + 1 ;
}
void Quick_sort (int arr[] , int low , int high)
{
    if (low < high)
    {
        int pi = partition (arr , low , high) ;
        Quick_sort (arr , low , pi - 1) ;
        Quick_sort (arr , pi+1 , high) ;
    }
}
void Quic_Sort (int arr[] , int n)
{
    step = 0 ;
    Quick_sort (arr , 0 , n - 1) ;
    for (int i=0 ; i<n ; i++)
    {
        printf("%d " , arr[i] ) ;
    }
    printf("\nThe step is : %d\n" , step) ;
}
int main()
{
```

```
48    int n = 9 , array1[9] , array2[9] , array3[9] ;
49    printf("Enter the 9 digit of my id number : ") ;
50    for (int i=0 ; i<n ; i++)
51    {
52        scanf("%d" , &array1[i] ) ;
53    }
54    for (int i=0 ; i<n ; i++)
55    {
56        array2[i] = array1[i] ;
57        array3[i] = array1[i] ;
58    }
59    for (int i=0 ; i < n-1 ; i++)
60    {
61        for (int j=i+1 ; j<n ; j++)
62        {
63            if (array2[i] > array2[j] )
64            {
65                int temp = array2[i] ;
66                array2[i] = array2[j] ;
67                array2[j] = temp ;
68            }
69        }
70    }
71    for (int i=0 ; i<n-1 ; i++)
```

```
    }
    for (int i=0 ; i<n-1 ; i++)
    {
        for (int j = i+1 ; j<n ; j++)
        {
            if ( array3[i] < array3[j] )
            {
                int temp = array3[i] ;
                array3[i] = array3[j] ;
                array3[j] = temp ;
            }
        }
    }
    printf ("My id is : ") ;
    Quic_Sort(array1 , n) ;
    printf ("Ascending order : ") ;
    Quic_Sort(array2 , n) ;
    printf ("Descending order : ") ;
    Quic_Sort(array3 , n) ;
    return 0 ;
}
```

OUTPUT

```
E:\iTubeGo\Code\Quic.exe

Enter the 9 digit of my id number : 2 4 2 0 1 4 1 6 5
My id is : 6 5 4 4 2 2 1 1 0
The step is : 36
Ascending order : 6 5 4 4 2 2 1 1 0
The step is : 55
Descending order : 6 5 4 4 2 2 1 1 0
The step is : 55

Process returned 0 (0x0)   execution time : 7.951 s
Press any key to continue.
```